25.03.2024 | Dynamic Modelling Exam Summary Lecture

Attendees: Students

WHY?

- Processes in the real world can be described and simulated by computers.
- One way of embedding intelligence in systems is using computer model that;
    - Understands the human behavior
    - Can mimic human behaviour
- Questions that we have answered:
    - How to ensure that a system "*understands*" your direction

WHAT?

- How to make dynamic computer models of process *(using excel and L2-Python)*

MODEL

- Our focus was on COMPUTATIONAL DYNAMIC MODELS:
    - Describes a specific phenomenon
    - Can simulate the process
- Important observations:
    - f

Users of Models (1st learning purpose)

- People:
    - Researchers
    - Policy makers
- Systems:
    - Adaptive systems
    - Human-like systems

Use cases of model validation

- During construction:
    - Test whether model fits expectations / reality
    - Results: is the model sufficiently correct?
- When using for understanding and studying a phenomenon:
    - Scenticis method: formulate hypothesis and test it
    - Result: better insight in process

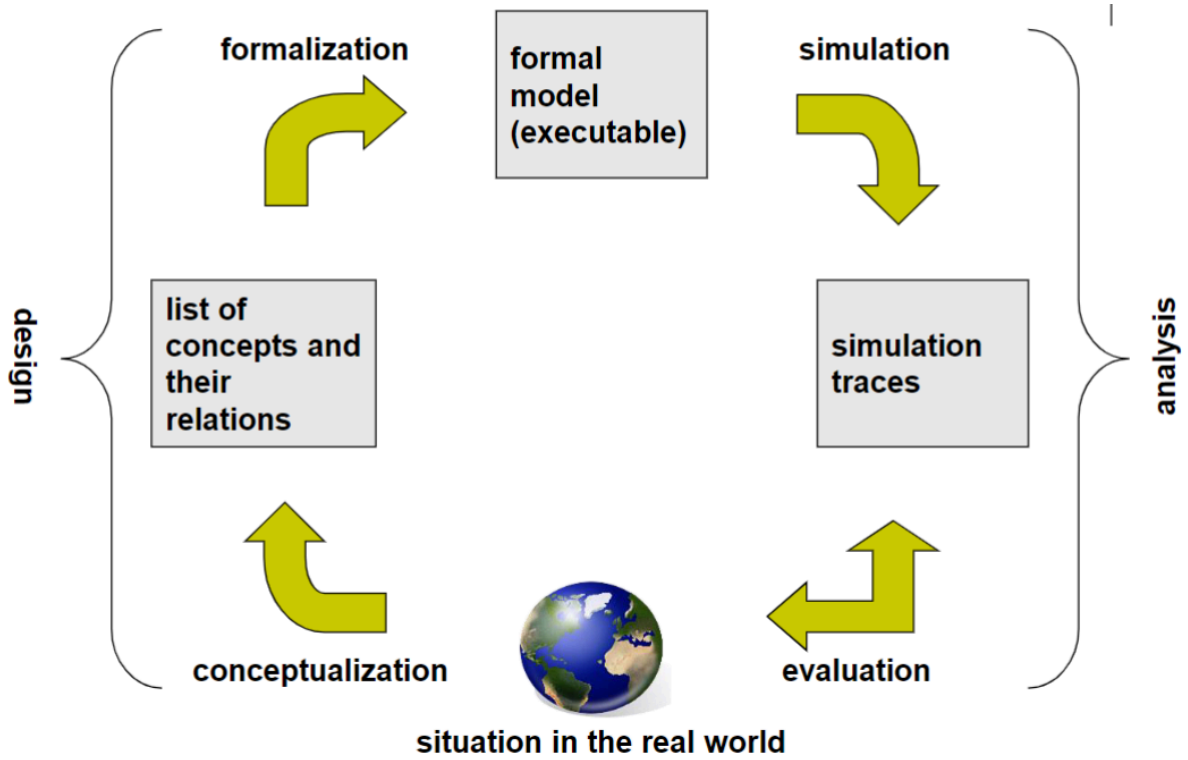
Properties of behavior of dynamic models

- Follow from hypothesis / research question
- Statements about:
    - Effects that happen over time
    - Combination of inputs that lead to some output
- Examples:
    - Jupiter and Pluto will not collide within two years
    - If the interest decreases, the long term effect on household income is positive
    - When taking a pull every 3 hours the medication level is between 12 and 20 mg/l

Validation of properties

- Run (*several*) simulations of the model and check the properties -> scenarios
    - Choose conditions / input
    - be precise about what you can conclude
- Alternatively:
    - use logic / mathematics to (dis)prove the properties

Modeling and simulation cycle:



Steps:

1) Conceptualisation: identification of important concepts and relations
2) Formalisation: describe concepts and relations in an unambiguous way
3) Simulation: running experiments with models to generate simulations traces
4) Evaluation: verify whether the simulations are in line with expectations (properties)

Numeric Representation:

- Concepts:
    - variables met numerical value
        - integer, real, 0 or 1
        - example: concept “temperature” → “Temp: real”

- Relations:
    - calculation rules to determine value of concept at next time point based on values other concepts
        - in dynamic model values change over time
        - t (change in time) is time step between two states

Logical representation:

- Concepts:
    - Statements (propositions) that can be true or false
        - Ex: at_location(food, p3)

- Relations:
    - Boolean rules that specify which combination of truth values is required for a concept to be true at next time point

(Semi-)formal notation:

- Numerical temporal rules:
    -
- Semi-formal logical temporal rules:

- Formal logical temporal rules:

Reporting

- Structured reporting is important
    - many arbitrary choices in process

- Elements of report:
    - assumptions, choices, motivation
    - concepts, relations, expected patterns
    - scenarios used in simulation experiments
        - Settings
        - input parameters
    - Verified properties, conclusions

The worst report of a model (simulation) ever:

- No clear research question
- Formal concepts do not match informal concepts
- Formal relations do not match informal concepts
- Specification does not run
- No evaluation
- No documentation of modeling choices
- Mix-up of different levels of aggregation
- Text is unreadable
- Handed in too late

Building blocks for models

- Different models have been discussed:
    - a

BUILDING BLOCKS - 1

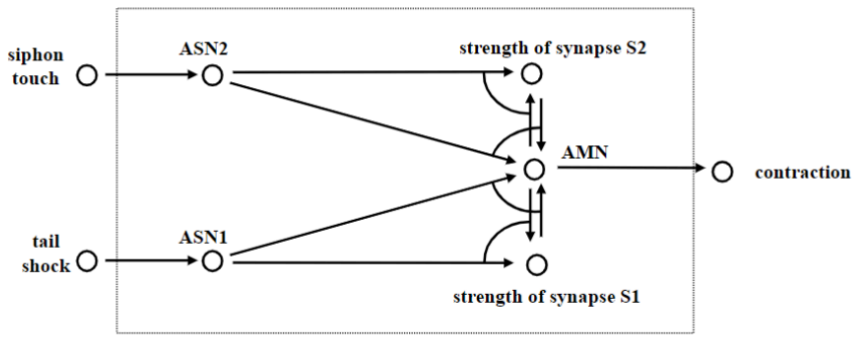
- Increase of factors
    - Growth:
        - linear
            - $Amount(t+\Delta t) = Amount(t) + c$
        - exponential
            - $Amount(t+\Delta t) = Amount(t) * \alpha$
        - logistic
            - $Amount(t+\Delta t) = Amount(t) * \alpha * ((C - Amount(t)) / C)$

BUILDING BLOCKS - 2

- Calculating with binary values in a numerical representation
  - AND
    - $C(t+\Delta t) = A(t) * B(t)$
  - NOT
    - $C(t+\Delta t) = (1 - B(t))$
  - OR
    - $C(t+\Delta t) = MAX(A(t), B(t))$

BUILDING BLOCKS - 3

- Learning / adaptation:
  - Adjustment of the synapse strength based on simultaneous activation of MN and SN2
    - $S2(t+\Delta t) = \begin{cases} S2(t) + 1 & \text{if } S2(t) < 4 \text{ and } AMN(t) = ASN2(t) = 1 \\ S2(t) & \text{else} \end{cases}$



BUILDING BLOCKS - 4

- Different type of behaviour:
  - Reactive
    - stimulus – response
    - delayed response
  - Motivation-based
    - via desires and intentions
  - Adaptive
    - via “sensitivity”
- Environmental complexity thesis:
  - The more complex the environment → the more sophisticated the behaviour that is required to deal with this environment → the more complex the mental representations and capabilities needed.

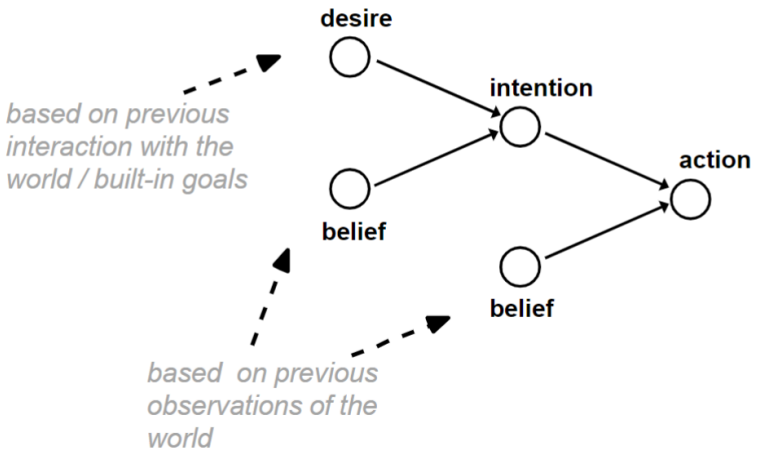
BUILDING BLOCKS - 5

- Vocabulary for behavioral & agent modeling
  - external perspective:
    - observations & actions
  - internal perspective:
    - Cognitions
      - beliefs, sensory_representations, preparation_states
- extended mind
  - no cognitions, only observations + actions
- Agent
  - internal concepts: world model, self model, history
    - desires, intentions, communications

concept	***predicate***
sensory_represenation	**sr : AGENT x OBSERVATION**
preparation_state	**ps : AGENT x ACTION**
belief	**belief : AGENT x INFO_ELEMENT**
desire	**desire : AGENT x INFO_ELEMENT**
passive observation	**observed : AGENT x INFO_ELEMENT**
active observation	**to_be_observed : AGENT x INFO_ELEMENT**
action	**performed : AGENT x ACTION**
incoming communication	**communicated_from_to : INFO_ELEMENT x AGENT x AGENT**
outgoing communication	**to_be_communicated_from_to : INFO_ELEMENT x AGENT x AGENT**

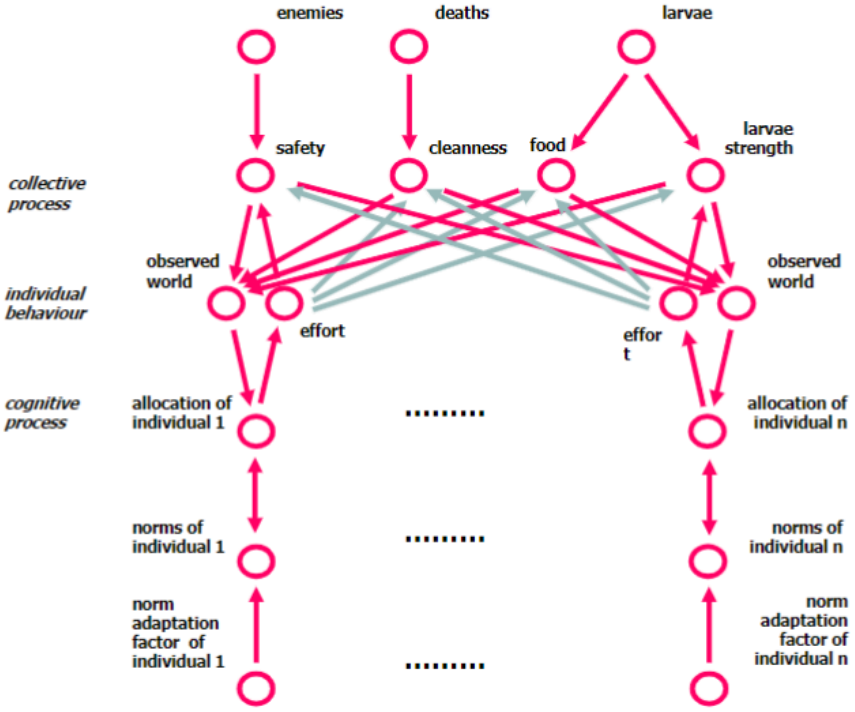
BUILDING BLOCKS - 6

- BDI model:



BUILDING BLOCKS - 7

- Integration of models of different levels
  - cognition – behaviour – collective



BUILDING BLOCKS - 8

- Two approaches:
  - Modeling groups
    - population-based (collective)
    - one concept for number of individuals in a category
    - model describes transitions between categories
    - abstracted but scalable
  - Modeling individuals
    - individual-based
    - one model per individual, individual behaviour
    - precise but not easy to scale

BUILDING BLOCKS - 9

- Population-based models usually require stochastic variables
  - precise factors have been abstracted

  InfectionStateA1(t+Δt) = 1 ***IF***
  InfectionStateA1(t) = 1 ***AND*** *r ≥ RecoveryRate*
  (where *r* is random number between 0 and 1)

- Stochastic models require several simulation runs to draw conclusions